MATH 708–2026. HOMEWORK 6. DUE ON MARCH 27.

- Unless otherwise stated, the exercises are from Voisin's textbook.
- The homework problems may be submitted orally (during assigned office hours), in written/typed form, or as a combination of the two.
- You may submit any subset of problems totaling up to 100 points.
- If you submit more than 100 points' worth, I will grade only the first 100 points you designate.
- Do not write anything on the cover sheet, as it will be used for grading. Bring it to office hours when you submit problems orally, and staple it to your written/typed solutions.

Problem 1 (10 points) Consider any resolution of an object F in an abelian category, $F \to A^\bullet$, and an injective resolution of an object G, $G \to I^\bullet$, together with a morphism $\alpha\colon F \to G$. Prove that there exists a morphism of complexes $\phi^\bullet\colon A^\bullet \to I^\bullet$ extending $\alpha\colon F \to G$.

In other words, prove that there exists a commutative diagram

$$\begin{tikzcd} F \arrow[r] \arrow[d, "\alpha"'] & A^0 \arrow[r] \arrow[d, "\phi^0"'] & A^1 \arrow[r] \arrow[d, "\phi^1"'] & A^2 \arrow[r] \arrow[d, "\phi^2"'] & \cdots \\ G \arrow[r] & I^0 \arrow[r] & I^1 \arrow[r] & I^2 \arrow[r] & \cdots \end{tikzcd}$$

Problem 2 (10 points) Given two morphisms of complexes, as in the previous problem, show that they are homotopic.

Problem 3 (10 points) Given a short exact sequence

$$0 \to F \to G \to H \to 0$$

in an abelian category with enough injective objects, and injective resolutions $F \to I^\bullet$ and $H \to K^\bullet$, prove that there exists an injective resolution $G \to J^\bullet$ such that every J^i is isomorphic to $I^i \oplus K^i$.

Furthermore, show that there exists a short exact sequence of complexes

$$0 \to I^\bullet \to J^\bullet \to K^\bullet \to 0$$

making the following diagram commutative:

$$\begin{tikzcd} 0 \arrow[r] & F \arrow[r] \arrow[d] & G \arrow[r] \arrow[d] & H \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & I^\bullet \arrow[r] & J^\bullet \arrow[r] & K^\bullet \arrow[r] & 0 \end{tikzcd}$$

Problem 4 (10 points) Show that every flasque sheaf $\mathcal{F}$ on a topological space X is Γ-acyclic.

Problem 5 (10 points) Consider a functor $F\colon \mathcal{A} \to \mathcal{B}$ between abelian categories. Show that if F has a right adjoint, then F is right exact, and if F has a left adjoint, then F is left exact.

Problem 6 (10 points) Let R be a commutative ring. For any R-module E, let $E^\vee = \operatorname{Hom}_{\mathbf{Z}}(E, \mathbf{Q}/\mathbf{Z})$.

(a) Show that
$$0 \to A \to B \to C \to 0$$
is exact if and only if
$$0 \to C^\vee \to B^\vee \to A^\vee \to 0$$
is exact.

(b) Let F be a flat R-module, and let I be an injective R-module. Show that $\operatorname{Hom}_R(F, I)$ is injective.

(c) Show that an R-module E is flat if and only if $E^\vee$ is injective.

Problem 7 (5 points) Let X be a connected topological space. Show that the constant sheaf with stalk $\mathbf{Z}$ is flasque if and only if X is an irreducible topological space, that is, if and only if X cannot be expressed as the union of two proper closed subsets.

Problem 8 (10 points) Is it true that the quotient of a flasque sheaf by a flasque subsheaf is flasque?

Problem 9 (10 points) Let $\mathcal{F}$ be a sheaf that is injective as an object in the category of sheaves. Show that $\mathcal{F}$ is flasque.

Problem 10 (10 points) Consider $X = \mathbf{CP}^1$ with the Zariski topology; that is, a subset is closed if and only if it is finite or equal to X. Let K be the field of rational functions in 1 variable, viewed as a constant sheaf on X.

(a) Show that X carries a structure sheaf $\mathcal{O}_X$ defined as follows: for each open set $U \subset X$, the ring $\mathcal{O}_X(U) \subset K$ consists of those rational functions having no poles on U.

(b) Show that $0 \to \mathcal{O}_X \to K \to K/\mathcal{O}_X \to 0$ is a flasque resolution of $\mathcal{O}_X$.

(c) Use part (b) to show that $H^i(X, \mathcal{O}_X) = 0$ for $i > 0$.

Problem 11 (10 points)

(a) Show that the pushforward of a flasque sheaf is flasque.

(b) Let $i\colon Y \hookrightarrow X$ be a closed subset of a topological space X, equipped with the induced topology, and let $\mathcal{F}$ be a sheaf on Y. Show that
$$H^k(Y, \mathcal{F}) \cong H^k(X, i_*\mathcal{F}).$$

Problem 12 (10 points) Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. We define $\operatorname{Ext}^i(\mathcal{F}, \cdot)$ as the right derived functors of $\operatorname{Hom}(\mathcal{F}, \cdot)$, and we define $\mathcal{E}xt^i(\mathcal{F}, \cdot)$ as the right derived functors of $\mathcal{H}om(\mathcal{F}, \cdot)$.

(a) Show that $\mathcal{E}xt^0(\mathcal{O}_X, \mathcal{G}) = \mathcal{G}$, $\mathcal{E}xt^i(\mathcal{O}_X, \mathcal{G}) = 0$ for $i > 0$.

(b) Show that $\operatorname{Ext}^i(\mathcal{O}_X, \mathcal{G}) \cong H^i(X, \mathcal{G})$.

(c) Let $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ be a short exact sequence of sheaves. Deduce the following long exact sequences:

$$\cdots \to \mathrm{Ext}^i(\mathcal{F}'', \mathcal{G}) \to \mathrm{Ext}^i(\mathcal{F}, \mathcal{G}) \to \mathrm{Ext}^i(\mathcal{F}', \mathcal{G}) \to \mathrm{Ext}^{i+1}(\mathcal{F}'', \mathcal{G}) \to \cdots$$

and

$$\cdots \to \mathcal{E}xt^i(\mathcal{F}'', \mathcal{G}) \to \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) \to \mathcal{E}xt^i(\mathcal{F}', \mathcal{G}) \to \mathcal{E}xt^{i+1}(\mathcal{F}'', \mathcal{G}) \to \cdots .$$

Problem 13 (10 points) Let G be a group, and let A be a G-module. We define the group cohomology

$$H^n(G, A) = \mathrm{Ext}^n_{\mathbf{Z}[G]}(\mathbf{Z}, A),$$

where $\mathbf{Z}$ is viewed as a trivial G-module.

(a) Show that $H^n(G, A)$ computes the right derived functors of the left-exact covariant functor $A \mapsto A^G$ from the category of G-modules to the category of abelian groups.

(b) Let $G = \mathbf{Z}$, and let $A = \mathbf{Z}$ be a trivial G-module. Compute $H^n(G, A)$.

Problem 14 (15 points) Let $\mathcal{F}$ be a sheaf on a topological space X. Consider the set of all open coverings of X as a partially ordered set, ordered by refinement.

(a) Show that the Čech cohomology groups $\{\check{H}^\bullet_{\mathcal{U}}(X, \mathcal{F})\}_{\mathcal{U}}$ form a direct system. Define $H^\bullet_{\text{Čech}}(X, \mathcal{F}) = \varinjlim_{\mathcal{U}} \check{H}^\bullet_{\mathcal{U}}(X, \mathcal{F})$.

(b) Show that $H^\bullet(X, \mathcal{F}) \cong H^\bullet_{\text{Čech}}(X, \mathcal{F})$.